



Name: Cohort/Batch: Date: Score:

LOAD BALANCER PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Networking

TOTAL: 10 QUESTIONS

Q1 What is Load Balancer and what problem in Networking does it solve?

Q6 How does Load Balancer handle reliability and high availability?

Q2 What are the core components or concepts in Load Balancer?

Q7 What observability does Load Balancer provide out of the box?

Q3 How does Load Balancer compare to its main alternatives?

Q8 What are common performance bottlenecks in Load Balancer?

Q4 How do you get started with Load Balancer for a new project?

Q9 What security best practices apply when running Load Balancer?

Q5 What configuration options most affect Load Balancer's behavior in production?

Q10 What mistakes do teams commonly make when adopting Load Balancer?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Load Balancer is a tool or platform

Mitigation

Load Balancer is a tool or platform that automates or simplifies a specific concern in the Networking domain, reducing manual effort and operational complexity for engineering teams.

A6 Load Balancer provides HA through leader election, Observability

Load Balancer provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Load Balancer has key building blocks that

Primitives

Load Balancer has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Load Balancer exposes metrics (Prometheus endpoint or Information

Load Balancer exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Load Balancer trades off simplicity against feature

Hardening

Load Balancer trades off simplicity against feature richness differently than its alternatives. Choose Load Balancer when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Load Balancer via its package manager

Gotchas

Install Load Balancer via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Load Balancer with the principle of

Assessment

Run Load Balancer with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency, Configuration

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and Organization

Organization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.